

dirs, etc) and subjects (processes) in your system. The user identity component of the Security context will have one of the above three identities and nothing else.

Similarly, to check the available roles, use the following code:

```
[root@vbg ~]# seinfo -r
```

```
Roles: 6
staff_r
user_r
object_r
secadm_r
sysadm_r
system_r
```

And please check the available list of types yourself by using the command given below:

```
[root@vbg ~]# seinfo -t
```

Thus we now know that any existing object or subject in the system will have a security context created out of these three users, six roles and 1,513 types.

You can alter the security context of any object by using the *chcon* command.

To change the type of an object, use *chcon -t*. To change the user identity, use *chcon -u* and *chcon -r* for role.

To test the above, create an empty file "context" in the */tmp* directory and check its default security context:

```
[root@vbg ~]# touch /tmp/context
[root@vbg ~]# ls -lZ /tmp/context
-rw-r--r-- root:root user_u:object_r:tmp_t:0 /tmp/context
```

To change the type of this object from *tmp_t* to *unconfined_t*, use the code below:

```
[root@vbg ~]# chcon -t unconfined_t /tmp/context
[root@vbg ~]# ls -lZ /tmp/context
-rw-r--r-- root:root root:object_r:unconfined_t:0 /tmp/context
```

I leave it as an exercise for you to change the role and the user for this object. If you face any issues, feel free to contact me at the e-mail ID provided below.

Assuming that, as above, you have changed the security context of an object quite a few times and you would like to revert back to the original/default security context, *restorecon* comes to the rescue.

restorecon restores files to their default security contexts. The verbose option of the *restorecon* command also displays the changes made in the security context. To check the usage of this very handy command, look at the example below:

```
[root@vbg ~]# touch /home/vbg/test-context
[root@vbg ~]# ls -lZ /home/vbg/test-context
```

```
-rw-rw-r-- vbg vbg
user_u:object_r:user_home_t:0 /home/vbg/test-context
```

```
[root@vbg ~]# chcon -t tmp_t /home/vbg/test-context
```

```
[root@vbg ~]# ls -lZ /home/vbg/test-context
-rw-rw-r-- vbg vbg user_u:object_r:tmp_t:0 /home/vbg/test-context
```

```
[root@vbg ~]# restorecon -v /home/vbg/test-context
restorecon reset /home/vbg/test-context context user_u:object_r:tmp_t:0-
>user_u:object_r:user_home_t:0
```

```
[root@vbg ~]# ls -lZ /home/vbg/test-context
-rw-rw-r-- vbg vbg user_u:object_r:user_home_t:0 /home/vbg/test-context
```

The above snippet shows the utility of the *restorecon* command. It reads the default contexts to be applied from the policy files and applies the default security context to a file/directory/object.

A word of caution when using *restorecon*: DO NOT use the *restorecon* command with the *-r* (recursive) option. It may overwrite the security contexts of some important files in the system that you may have changed.

If you think that you have spoiled the security contexts of the files in your system beyond recovery, do not panic. Help is available in the form of Auto-Relabel, at boot. Simply create an empty file */.autorelabel*. Please note that it is a hidden file.

```
[root@vbg ~]# touch /.autorelabel
[root@vbg ~]# reboot
```

Following the procedure mentioned above will cause SELinux to relabel the files on your system upon rebooting. Please use this to fix any improper security context on files and directories.

You can also use the *fixfiles* command to achieve the above. *fixfiles* can prevent you from rebooting your system but may not be as effective. Depending on the options and time available, you can choose any option that suits you—though I would suggest the reboot option.

Still to come

So, that's about it for now. I'll be back the following months with topics such as:

- Understanding the Targeted Policy - Part II
- Policy Modules
- MLS and MCS 

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